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**SARDAR PATEL UNIVERSITY**

**M.Sc. (Information Technology) – Semester- I NC Examinations-2025**

## PS01CINT52 – Python Programming

**Friday, 11<sup>th</sup> April 2025**

**Time: 10.00 AM - 01.00 PM**

**Max. Marks: 70**

**Q-1 Select the appropriate answer for the following questions:**

[ 08 ]

- i. Python files are saved with the extension \_\_\_\_\_.
  - A. .pe
  - B. .pi
  - C. .py
  - D. .python
- ii. Comments in Python begin with \_\_\_\_\_.
  - A. &
  - B. \*
  - C. %
  - D. #
- iii. Which of the following is immutable (values that cannot be changed)?
  - A. List
  - B. Dictionary
  - C. Tuple
  - D. Set
- iv. What will be the output after the following statements?

```
x = 16 / 4 * 5
y = 16 / 4 * 5.0
z = 16 / 4.0 * 5
print(x, y, z)
```

  - A. 25 15 20
  - B. 20.0 20.0 20.0
  - C. 20.0 20 20.0
  - D. 20 20.0 20
- v. What will be the data type of p after the following statement, if input entered is 51?

```
p = input('Please enter a number: ')
```

  - A. Integer
  - B. Float
  - C. String
  - D. List
- vi. What is the default value of the step value in the function range()?
  - A. 1
  - B. -1
  - C. 0
  - D. None of these
- vii. The `def` keyword is used in Python to \_\_\_\_\_.
  - A. Define a list
  - B. Define an array
  - C. Define a function
  - D. Define a dictionary
- viii. \_\_\_\_\_ is used for developing GUI Application in Python.
  - A. Tkinter
  - B. Tinker
  - C. Kinter
  - D. None of these

CP.T02

**Q-2 Give answer in short. ( ANY SEVEN )**

**[ 14 ]**

- i. Write down the advantages of Python programming language.
- ii. List out different data types provided by Python programming language.
- iii. Write the differences between Local and global variable in Python.
- iv. Explain in brief, range() function.
- v. List out any six built-in functions available in Python.
- vi. Write the differences between mutable and immutable data types in python.
- vii. Differentiate between Tuple and List.
- viii. Explain the concept of type casting with example.
- ix. List out modes of opening the file with their meaning.

**Q-3 A. Explain input() and print() functions with examples.**

**[ 6 ]**

**B. Explain List in detail.**

**[ 6 ]**

**OR**

**B. Explain Tuple in detail.**

**[ 6 ]**

**Q-4 A. List out different types of operators available in Python. Explain any one type of operators in detail.**

**[ 6 ]**

**B. What is a string? Explain any three built-in String methods with example.**

**[ 6 ]**

**OR**

**B. What do you mean by loop? List out the looping structure available in Python. Explain any one of them in detail.**

**[ 6 ]**

**Q-5 A. Explain (i) How to define a function? (ii) How to call a function? (iii) Return statement.**

**[ 6 ]**

**B. What do you mean by module? Discuss Python modules with example.**

**[ 6 ]**

**OR**

**B. What is inheritance? List different types of inheritance. Explain any one of them.**

**[ 6 ]**

**Q-6 A. Write a note on Exception handling mechanism in Python.**

**[ 6 ]**

**B. Discuss File handling in Python.**

**[ 6 ]**

**OR**

**B. Explain database connectivity in Python with examples.**

**[ 6 ]**

